

The Utility Problem

Electric utilities are required to measure and report distribution reliability performance to regulators using standardized indices — most commonly SAIFI (System Average Interruption Frequency Index), SAIDI (System Average Interruption Duration Index), CAIDI (Customer Average Interruption Duration Index), CEMI (Customers Experiencing Multiple Interruptions), and CELID (Customers Experiencing Long Interruption Durations). These indices are not simply outage counts or event logs — they are normalized, customer-weighted metrics computed from outage data, designed to represent the reliability experience of the average customer on the system.

While most utilities calculate these indices at the system level for regulatory reporting, the indices are far more valuable when computed at the feeder (circuit) level. A system-level SAIFI that appears acceptable may mask one or two chronically underperforming feeders that are responsible for the majority of customer interruptions. Without feeder-level disaggregation, operations and planning teams have no systematic visibility into which circuits are driving reliability degradation, making it difficult to prioritize capital investment, maintenance programs, or targeted hardening projects.

Compounding this, reliability indices are rarely tracked as time series. Most utilities produce annual regulatory submissions but do not maintain multi-year trending of feeder-level indices, making it difficult to determine whether a particular circuit is improving, deteriorating, or seasonal in its reliability behavior. Benchmarking against regulatory targets or peer feeders is similarly ad hoc. The result is that reliability management remains largely reactive — driven by individual outage events rather than by systematic, data-driven identification of the circuits that most need intervention.

The Data Solutions and Methods

The Awesense Platform ingests outage event data from the utility's Outage Management System (OMS) and normalizes it against the customer and feeder topology data held in the Awesense Energy Data Model (EDM). The platform exposes this unified dataset through its APIs, giving external analytical tools direct, structured access to all outage, customer, and feeder data required to compute reliability indices. The index calculation itself — SAIFI, SAIDI, CAIDI, CEMI, and CELID at the feeder level — might be implemented as an analytical layer outside the platform, for example, as a Python-based processing pipeline that queries the Awesense APIs, applies the IEEE 1366¹ index formulas per feeder and reporting period, and writes results back for storage and visualization.

¹ IEEE 1366 acts as the industry standard for defining and calculating electric power distribution reliability indices, including SAIDI and SAIFI. It provides utilities with a consistent, standardized framework to measure system performance and benchmark service quality.

Reliability Metrics Trending and Benchmarking

[UC57] - Awesense Platform Use Case

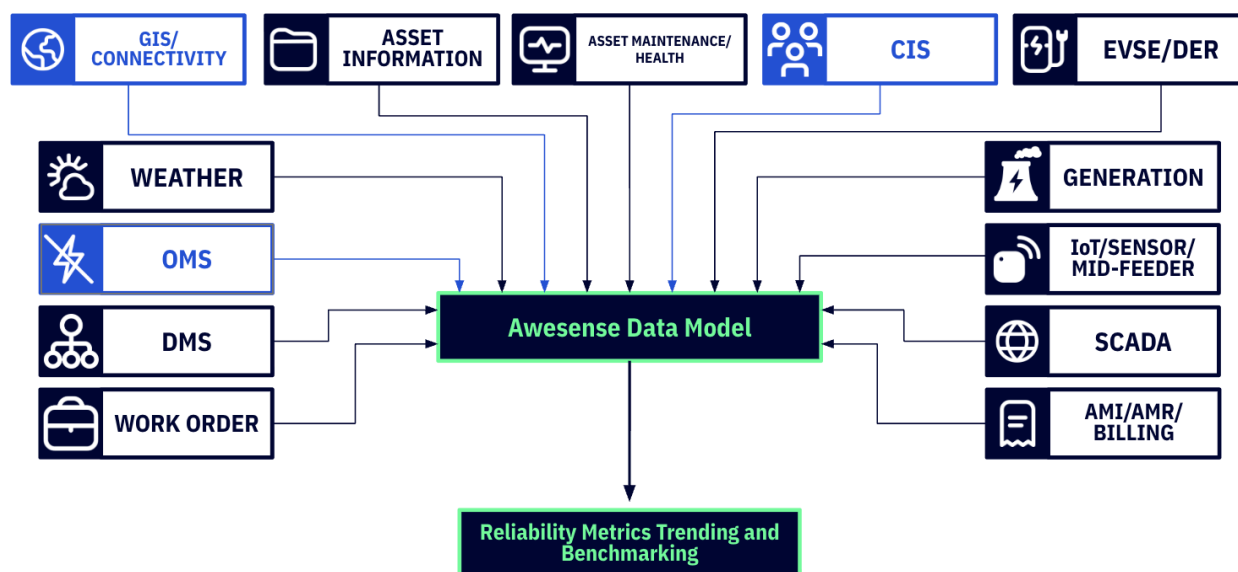


This approach produces a longitudinal performance record for every circuit in the service territory, computed monthly, quarterly, and annually across multiple years of historical data. These time series form the foundation for two complementary analytical layers: trending and benchmarking. Trending analysis identifies whether a feeder's indices are improving or deteriorating over time, detects seasonal reliability patterns, and flags circuits whose metrics have crossed a defined threshold or shown statistically significant deterioration across recent periods. Benchmarking compares each feeder's computed indices against regulatory targets, internal performance standards, or the distribution of indices across the full feeder fleet — surfacing circuits that are clear outliers relative to peers.

Geospatial heatmap visualization is a key output of the feeder-level analysis. True Grid Intelligence (TGI) is a web-based digital twin explorer that provides a visual interface to the data stored in the Awesense Platform. The TGI Segment Analyzer component enables users to create polygon overlays over feeders and geographic areas and apply heatmapping to any computed metric. Feeder-level SAIDI or SAIFI values are mapped onto their corresponding feeder polygons and rendered as a continuous heatmap across the service territory, giving operations and planning teams an immediate geographic view of where reliability performance is strongest and where it is most degraded — an effective format for communicating reliability risk to asset management and executive stakeholders.

Required Data Sets

The required data for this use case are visualized in the following diagram. Data sources highlighted in [royal blue](#) are required for this use case.



Available Use Case Implementations:

Users of the Awesense Platform can access this use case by the following means:

1. Please contact [Awesense](#) if you are interested in the implementation of this use case.